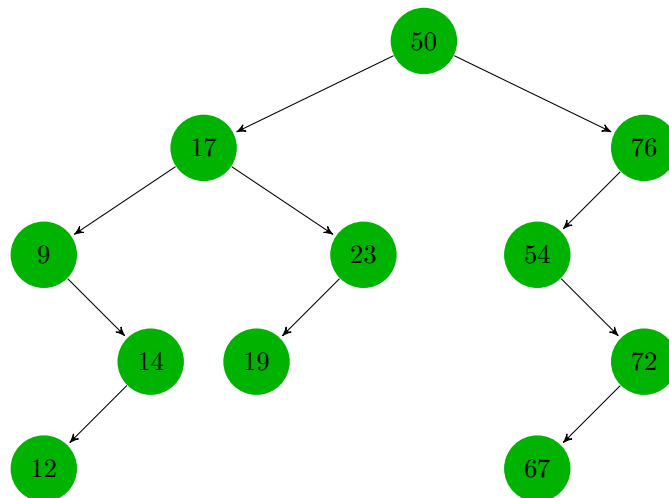


Weekly Assignment 12: Search Trees

1. (a) (10 points) How many binary search trees can you build with the following elements: $\{3, 5, 8, 12\}$? Draw them.
 (b) (10 points) How many AVL trees are there with the same elements $\{3, 5, 8, 12\}$? Draw them (or better, just point them out in your answer to the previous question).
2. (15 points) Consider the following binary search tree:



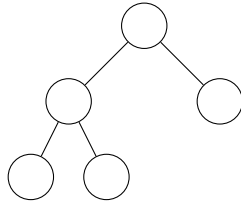
Show how this tree can be turned into an AVL tree using four rotations. Draw the binary search tree obtained after each rotation.

3. (15 points) Let us consider all the integers between 1 and $2^n - 1$. Give
 - (a) one insertion order of these integers maximizing for binary search trees height.
 - (b) one insertion order of these integers minimizing for binary search trees height (also giving a balanced BST).
4. (25 points) We extend the binary search tree data structure as follows: each node x has an additional field $x.lnodes$, storing the number of nodes in its left subtree.
 - (a) Write an algorithm for the following problem on such a tree: given k , find the node that ends up at index k of the in-order traversal of the tree. Your algorithm should run in $\mathcal{O}(h)$, where h is the height of the tree.
 - (b) Give the algorithm for insertion on our trees with the additional field. It should take $\mathcal{O}(h)$ time. Explain your answer.

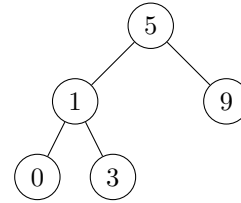
5. (25 points) You are working for *BST inc.*, an international company that delivers binary search trees as a service (BAAS). The main data centre of BST inc. got corrupted: all key values have been separated from their nodes. Fortunately, we still have the values themselves, and the underlying binary trees. It is your task to restore these.

Concretely, you are asked to develop an algorithm for the following problem. Given a binary tree T with n nodes, and a sorted integer array A of length n , assign key values to the nodes of T such that it becomes a binary search tree which has exactly the same values as A .

For instance, if the input is the array and binary tree on the left in the picture below, the result is the binary search tree on the right.



(a) Input array: $[0, 1, 3, 5, 9]$



(b) Output: result of filling in array values

Describe an efficient algorithm for this problem, together with a complexity analysis and an explanation why your algorithm is correct.